

WEEK 3 REPORT

Superstore Sales and Profit Analysis Using Power BI

1. Problem Statement

The project analyzes Superstore sales, profit, products, customers, regions, and order dates using Power BI. The aim is to identify sales trends and compare category and regional performance.

2. Objectives

Import and clean Superstore data.

Create a Date Table and hierarchy.

Establish relationships between tables.

Create DAX measures.

Analyze sales and profit trends.

Build an interactive dashboard.

3. Dataset

The Superstore dataset contains information about customer orders, products, sales and profitability.

Column	Description
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Customer ID	Unique customer identification
Customer Name	Name of the customer
Product ID	Unique product identification
Product Name	Name of the product
Category	Product category
Sub-Category	Product sub-category

Segment	Customer segment
Region	Geographical region
Sales	Sales amount
Quantity	Number of products ordered
Profit	Profit generated from the order

The dataset was used to analyze overall business performance and identify important sales and profit trends.

4. Power Query

Power Query was used to clean and transform the data. Data types were checked, and missing values, incorrect values, and duplicates were reviewed.

Order Date → Date

Ship Date → Date

Customer ID → Text

Product ID → Text

Category → Text

Sub-Category → Text

Segment → Text

Region → Text

Sales → Decimal Number

Quantity → Whole Number

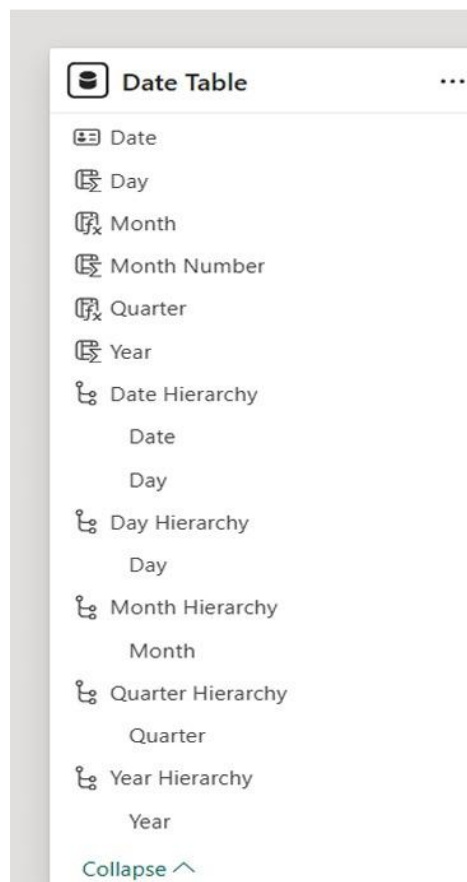
Profit → Decimal Number

Power Query was also used to check missing values, incorrect data types and duplicate records.

	Date	Year	Quarter	Month	Month Number	Day	Day of Week
1	20-09-2013	2013	3	9	9	20	Friday
2	21-09-2013	2013	3	9	10	21	Saturday
3	22-09-2013	2013	3	9	11	22	Sunday
4	23-09-2013	2013	3	9	12	23	Monday
5	24-09-2013	2013	3	9	13	24	Tuesday
6	25-09-2013	2013	3	9	14	25	Wednesday
7	26-09-2013	2013	3	9	15	26	Thursday
8	27-09-2013	2013	3	9	16	27	Friday
9	28-09-2013	2013	3	9	17	28	Saturday
10	29-09-2013	2013	3	9	18	29	Sunday
11	30-09-2013	2013	3	9	19	30	Monday
12	01-10-2013	2013	4	10	1	1	Tuesday
13	02-10-2013	2013	4	10	2	2	Wednesday
14	03-10-2013	2013	4	10	3	3	Thursday
15	04-10-2013	2013	4	10	4	4	Friday
16	05-10-2013	2013	4	10	5	5	Saturday
17	06-10-2013	2013	4	10	6	6	Sunday
18	07-10-2013	2013	4	10	7	7	Monday
19	08-10-2013	2013	4	10	8	8	Tuesday
20	09-10-2013	2013	4	10	9	9	Wednesday
21	10-10-2013	2013	4	10	10	10	Thursday
22	11-10-2013	2013	4	10	11	11	Friday
23	12-10-2013	2013	4	10	12	12	Saturday

5. Date Table

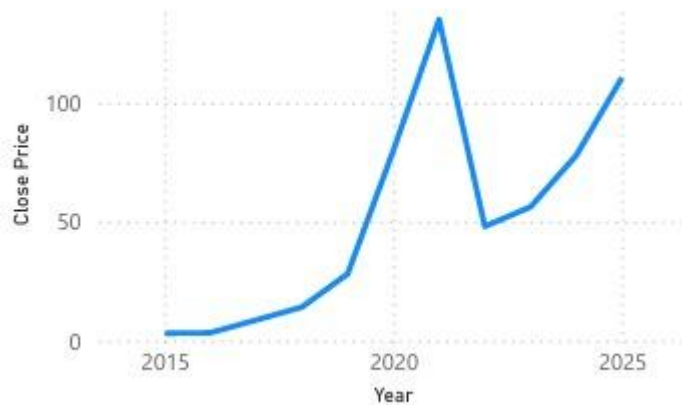
A Date Table was created with Date, Year, Quarter, Month, Month Number, Day, and Day of Week for time-based analysis.



6. Date Hierarchy

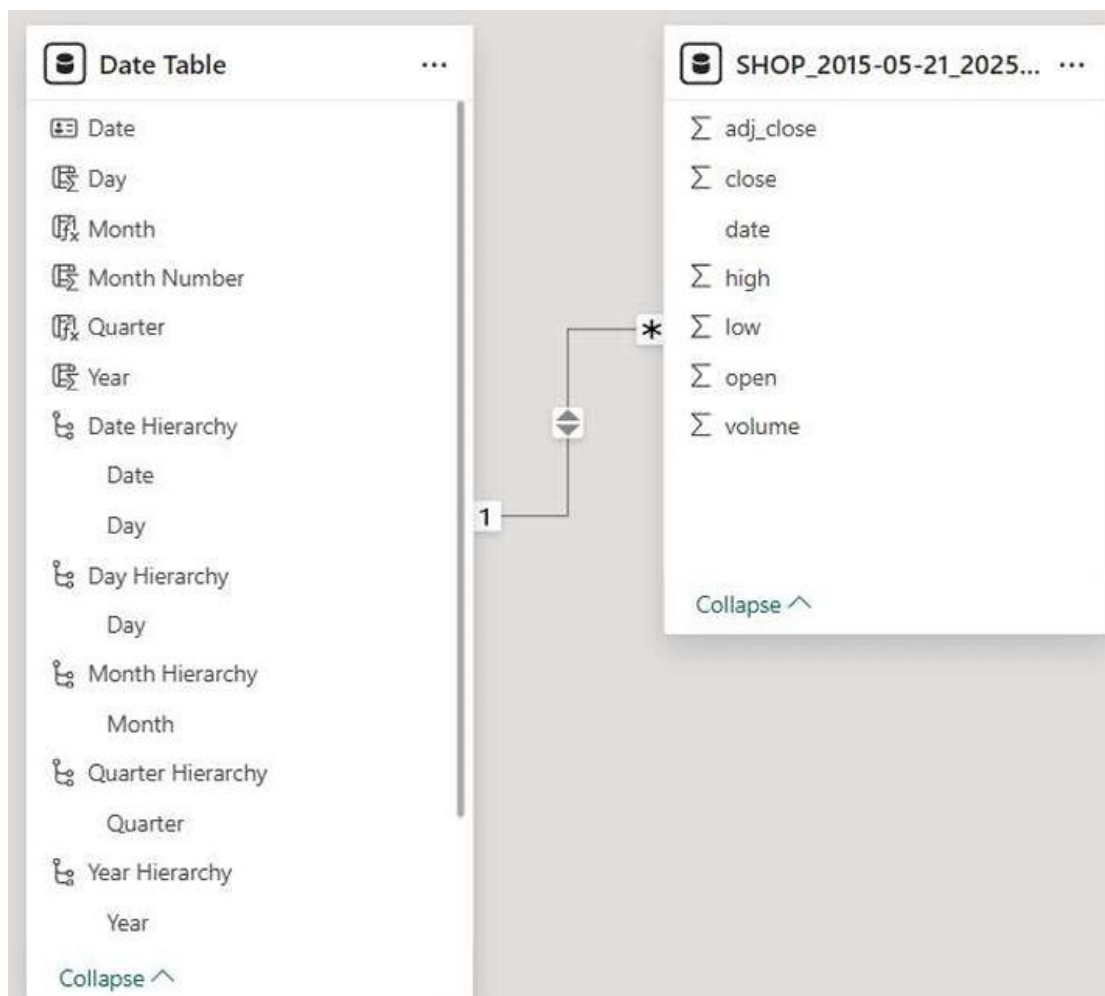
Year → Quarter → Month → Day was created for drill-down analysis.

Close Price by Year



7. Relationship

A one-to-many relationship was created between the Date Table and Sales table using Date → Order Date.



8. DAX Measures

Total Sales

Total Profit

Total Orders

Total Quantity

Average Sales

Profit Margin %

DAX Measures and Calculations

Total Sales

Total Sales = SUM(Superstore[Sales])

Total Profit

Total Profit = SUM(Superstore[Profit])

Total Quantity

Total Quantity = SUM(Superstore[Quantity])

Total Orders

Total Orders = DISTINCTCOUNT(Superstore[Order ID])

Average Sales

Average Sales = AVERAGE(Superstore[Sales])

Profit Margin %

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

These measures help analyze overall business performance.

9. Time Intelligence

Functions such as DATEADD, TOTALYTD, and SAMEPERIODLASTYEAR were used to compare performance across different periods.

Function	Purpose
DATEADD	Moves the date context to another period
TOTALYTD	Calculates year-to-date values
SAMEPERIODLASTYEAR	Compares with the previous year

These calculations help analyze changes in sales and profit over time.

10. Dashboard

An interactive **Superstore Sales and Profit Analysis Dashboard** was created Power BI

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Sales
- Profit Margin%

Slicers

- Year
- Category
- Region
- Segment

Visualizations

- Category-wise-sales - column chart
- Region-wise-profit - column chart
- Sales over time - line chart
- Sales and profit Analysis
- Decomposition tree - sales by category
- KPI cards - overall performance

The dashboard allows users to filter and analyze sales and profit on different years, categories, regions, and customer segments

11. Insights

cards show overall performance.

Category analysis shows sales contribution.

KPI Region analysis shows profit performance.

Sales trends can be viewed over time.

Slicers provide interactive filtering.

12. Conclusion

The project successfully used Power BI to clean, analyze, and visualize Superstore sales and profit data. DAX measures, time intelligence, charts, slicers, and a decomposition tree were used to create an interactive dashboard.

